

SUMMARY

This topic covers advanced number systems and series analysis techniques tailored for competitive General Aptitude examinations. It focuses on factorization mechanics, progression models (AP, GP, HP), and advanced series resolution methods such as telescopic and arithmetic–geometric progressions. Emphasis is placed on structured decoding strategies, prime factorization utilities, unit digit cycles, and rapid estimation frameworks for efficient exam performance.

This video provides a technical synthesis of advanced mathematical concepts in number systems and sequence decoding. Core areas include prime factorization for evaluating factor properties, progression-based modeling using common difference and common ratio, and structured approaches to solving complex series. The document also highlights high-speed problem-solving tools such as unit digit cyclicity, nested radical evaluation, and summation identities. The goal is conceptual clarity combined with fast execution.

1. FACTORIZATION AND PRIME DECOMPOSITION

Factorization involves expressing a composite number as a product of its prime components. Prime factorization forms the basis for determining number of factors, sum of factors, and product of factors.

1.1 FUNDAMENTAL PROPERTIES OF FACTORS

If a number N is written in prime factorized form as

$$N = a^p \cdot b^q \cdot c^r$$

where a, b, c are primes and p, q, r are their exponents, then:

Total Number of Factors (f):

$$(p + 1)(q + 1)(r + 1)$$

Sum of Factors:

$$[(a^{p+1} - 1)/(a - 1)] \times [(b^{q+1} - 1)/(b - 1)] \times [(c^{r+1} - 1)/(c - 1)]$$

Product of Factors:

$$N^{(f/2)}$$

These formulas allow rapid evaluation of factor-based properties without listing factors explicitly.

1.2 PERFECT CUBE AND PERFECT SQUARE CONSTRAINTS

For a number to be:

Perfect Square: Each prime exponent must be a multiple of 2.

Perfect Cube: Each prime exponent must be a multiple of 3.

Example:

$$162 = 2^1 \cdot 3^4$$

To convert into a perfect cube, adjust exponents to nearest multiples of 3:

$$\text{Multiply by } 2^2 \cdot 3^2 = 36$$

Result:

$$2^3 \cdot 3^6 \text{ (all exponents multiples of 3)}$$

2. NUMBER SERIES AND SEQUENCE DECODING

Missing number problems require analyzing patterns either through differences or ratios.

2.1 PRIMARY DECODING METHODOLOGIES

Common Difference (CD):

Used when terms change linearly at a steady rate.

Common Ratio (CR):

Used when growth is exponential or involves decimals.

Ratios may increase gradually (e.g., 1.5, 2.0, 2.5, 3.0).

Zigzag/Alternate Series:

Two interwoven sequences. Analyze alternate terms separately.

2.2 SPECIALIZED PATTERNS

Polynomial Series:

Patterns such as $n^2 + 1$, $n^2 + n$.

Prime Power Series:

Cubes or powers of prime numbers (e.g., 7^3 , 11^3 , 13^3 , 17^3).

Averaging Diagrams:

Central value equals $(\text{Left} + \text{Right})/2$ in structured diagrams.

Alpha-Numeric Series:

Map alphabets to numerical positions (A = 1, Z = 26).

Example: W = 23.

3. MATHEMATICAL PROGRESSIONS

Progressions follow structured term-generation rules.

3.1 ARITHMETIC PROGRESSION (AP)

Common difference: d

n-th Term (T_n):

$$a + (n - 1)d$$

Sum of n Terms (S_n):

$$n/2 (a + l)$$

or

$$n/2 [2a + (n - 1)d]$$

Arithmetic Mean between a and b:

$$(a + b)/2$$

3.2 GEOMETRIC PROGRESSION (GP)

Common ratio: r

n-th Term (T_n):

$$ar^{(n - 1)}$$

Sum of n Terms (S_n):

$$a(r^n - 1)/(r - 1)$$

Sum of Infinite GP (S_∞):

$$a/(1 - r), \text{ valid only if } |r| < 1$$

3.3 HARMONIC PROGRESSION (HP)

A sequence is in HP if its reciprocals form an AP.

Solution approach:

1. Convert terms to reciprocals.
2. Solve as AP.
3. Take reciprocal of final result if required.

4. ADVANCED SERIES RESOLUTION TECHNIQUES

4.1 TELESCOPIC SERIES

A telescopic series collapses due to cancellation of intermediate terms.

Standard Identity:

$$1/[n(n + 1)] = 1/n - 1/(n + 1)$$

General Rule:

If denominator factors differ by k , then:

$$\text{Sum} = (1/k) \times (\text{First Term} - \text{Last Term})$$

This method significantly reduces computation time.

4.2 ARITHMETIC–GEOMETRIC PROGRESSION (AGP)

An AGP consists of terms formed by multiplying an AP and a GP.

Example structure:

$2/2, 5/4, 8/8, 11/16 \dots$

Resolution Strategy:

1. Let total sum = S.
2. Multiply entire series by common ratio r.
3. Subtract rS from S.
4. AP component gets eliminated.
5. Solve resulting GP expression.

This converts AGP into a solvable infinite GP.

5. STRATEGIC SHORTCUTS AND ESTIMATION

5.1 UNIT DIGIT ANALYSIS

Unit digits follow cyclical patterns.

Example:

7^n cycle: 7, 9, 3, 1 (cycle length = 4)

Procedure:

1. Divide exponent by 4.
2. Use remainder to determine unit digit.

This avoids full power calculation.

5.2 NESTED RADICALS (INFINITE SQUARE ROOTS)

For expressions like:

$$\sqrt{x \pm \sqrt{x \pm \sqrt{x \dots}}}$$

Step:

1. Factor x into two consecutive numbers n and (n + 1).

If operation is addition (+):

Result = larger factor (n + 1)

If operation is subtraction (-):
Result = smaller factor (n)

This method provides instant evaluation.

5.3 STANDARD SUMMATION FORMULAS

Sum of first n Natural Numbers:
 $n(n + 1)/2$

Sum of first n Squares:
 $n(n + 1)(2n + 1)/6$

Sum of first n Cubes:
 $[n(n + 1)/2]^2$

Sum of first n Odd Numbers:
 n^2

Sum of first n Even Numbers:
 $n^2 + n$